Effects of Fiscal Shocks in a Globalized World. A Replication of Auerbach and Gorodnichenko (2016)*

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Abstract

This paper presents the results of a replication of Auerbach and Gorodnichenko (2016) based on the supplementary material provided by the authors. Using data on announced fiscal spending at a high-frequency, Auerbach and Gorodnichenko (2016) are able to resolve the exchange rate puzzle found in the empirical fiscal policy literature and offer a novel approach to fiscal shock identification. I show that the authors' replication files do not generate the exact results of the paper. While the main effect of government spending shocks on the exchange rate, an appreciation, is preserved, it is quantitatively smaller and lacks statistical significance. I discuss potential reasons for this finding. I also extend the sample period and show that the higher power is insufficient to restore significance. This paper aims to take the learnings from this replication and add to the discussion of reproducibility and replicability in economics.

Keywords: Fiscal Policy; Government Spending Shocks; Time Series Analysis; Replication;
JEL codes: C32: C82; E62; F41

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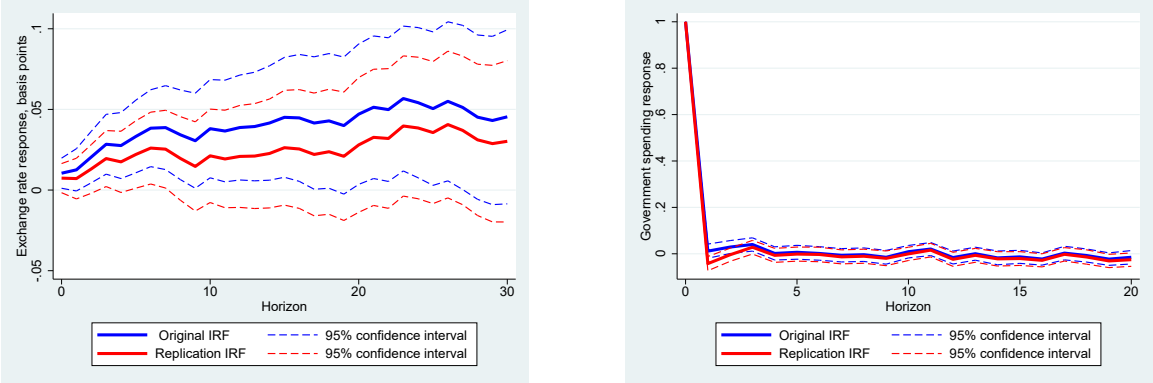
1 Introduction

This paper presents the results of a replication of a paper by Auerbach and Gorodnichenko (2016), henceforth AG, with the title "Effects of Fiscal Shocks in a Globalized World", which was published in the IMF Economic Review in 2016. AG show that using data on announced fiscal spending at a high frequency leads to an appreciation of the exchange rate and therefore resolves the so-called exchange rate puzzle previously found in the empirical literature (e.g. Kim and Roubini 2008; Monacelli and Perotti 2010). The novelty of their approach is the use of spending data at a daily frequency (generally lower frequency fiscal data is used for empirical analyses).

AG's approach is interesting for several reasons. Many papers have used identification methods based on quarterly data, accounting for anticipation in various ways. In comparison, high-frequency identification is relatively rare, but it adds many years of additional data. Moreover, the approach seems highly interesting for evaluating the response of other high-frequency variables, like financial variables, for instance. Furthermore, AG provide supporting evidence for the relevancy of foresight with respect to fiscal spending shocks. They construct two measures of government spending for their analysis: one accounting for anticipatory effects of spending announcements, and another one using actual spending. Both series are informative in the sense that while the former explicitly accounts for legislative and implementation lags and measures the impact of spending announcements on agents' expectations, the latter captures the actual financial impulse as if it came as a full surprise. AG show results for both series throughout.

When reproducing AG's results based on the supplementary material provided, I obtain the same direction of effects but of smaller magnitude than in the original publication. Hence, I am unable to exactly reproduce AG's tables and figures. I explore this difference further with respect to the main finding of AG: fiscal shocks based on daily spending announcements lead to an appreciation of the real exchange rate, thereby resolving the empirical exchange rate puzzle. While the standard errors are similar to those of AG, the smaller size of the estimates

leaves them statistically insignificant. Figure 1 illustrates the original and replicated impulse responses of the exchange rate measure to a fiscal shock, and illustrates these issues.



(a) Real Exchange Rate **(b)** Government spending

Figure 1: Impulse response functions (IRFs) to a unit shock in the DoD announcement government spending series for (a) the U.S. real exchange rate on the left and (b) government spending on the right. The x-axis (horizon) displays days. Red lines show the results from my replication, blue lines display the original results by AG.

I extend the spending series with an additional 40 644 daily observations of Department of Defense (DoD) spending announcements from 2014 to 2022 in order to boost the statistical power of the original findings. While the direction of the effects remains the same, more data does not restore significance. I identify the problems with reproducibility to be caused by inconsistencies in the files provided to deseasonalize the spending series.

These results contribute to a more general discussion on reproducibility in economics. They blend in well with findings by Gertler, Galiani, and Romero (2018) who find that when trying to reproduce figures and tables from 203 studies with provided materials (the selected articles were all published in May 2016 in nine top economic journals), only 14 percent can be fully replicated when starting from the raw data files. Berry, Coffman, Hanley, Gihleb, and Wilson (2017) examine how often replications actually happen, defining a replication "[...] as any project that reports results that speak directly to the veracity of the original paper's main hypothesis" (Berry et al. (2017, p.27)). I use this relatively broad definition in this paper as

well.[1] Based on the 100th volume of the American Economic Review (AER), the authors find that slightly less than 30 percent of the respective papers had been replicated. When including studies that extend the original articles, this number increases to 59 percent. Berry et al. (2017) also point out that most researchers are not well aware of existing replications and replication attempts (based on a survey conducted with the authors of the original articles and citing authors). Along those lines, Ankel-Peters, Fiala, and Neubauer (2023) review the impact of comments published in the AER between 2010 and 2020 based on the number of citations. They find comments to be rarely cited and without an influence on the original publication's number of citations. Similarly, Mueller-Langer, Fecher, Harhoff, and Wagner (2019) find that considering the Top 50 economics journals, only 0.1 percent of the articles published in the respective journals were replications. They point out that it is mostly highly cited and high-impact articles that are replicated. Being published in the Top 5 journals makes it less likely for an article to be replicated, as researchers expect those journals to be stricter and more selective compared to other journals. Similar exercises were also conducted for publications in development economics (Sukhtankar (2017)), in labor economics (Hamermesh (2007)) and for studies in experimental economics (Camerer, Dreber, Forsell, Ho, Huber, Johannesson, Kirchler, Almenberg, Altmejd, Chan, Heikensten, Holzmeister, Imai, Isaksson, Nave, Pfeiffer, Razen, and Wu (2016)), all coming to similar conclusions. The latter study also analyses how easily articles can be replicated. Camerer et al. (2016) are able to replicate 11 out of 18 studies which were published in the AER and QJE between 2011 and 2014. Furthermore, they conclude that while the sign and direction of the replicated effects is generally the same as the original, the effects are normally smaller (66 percent of the original). This summary is not exhaustive.[2] It shows that even though all major journals

[1]This is a very broad definition of the term replication. Other definitions rightfully distinguish between reproducibility and replicability (Vilhuber (2020) or Kirkby (2023), for instance), where the former term encompasses a confirmation of the original results based on the same data and code and the latter involves new statistical methods, code or data. According to this definition my analysis consists of both, reproduction and replication (especially through the extension). To improve readability I do not consider this distinction for my analysis.

[2]A good overview on the history and current condition of reproducibility and replicability in economics can also be found in Vilhuber (2020).

in economics have adopted a Data Access Policy requiring authors to make their code and data available for some years now, replication and reproduction studies still do not always run smoothly. The merit of those attempts is relatively straight forward: to provide further robustness and reliability. The aim of the current paper is explicitly not a challenge of the original publication but to illustrate the problems that can arise while replicating. AG serve as an example. The choice of this specific publication is driven by my interest in their work and the novelty and appeal of their approach.

The structure of this paper is as follows. Section 2 presents more information on the original paper by AG and places it in the literature. Section 3 details the results from my replication based on the authors' supplementary material, while Section 4 extends the original sample period and presents the corresponding results and Section 5 concludes.

2 Auerbach and Gorodnichenko (2016): "Effects of Fiscal Shocks in a Globalized World"

In their paper "Effects of Fiscal Shocks in a Globalized World", AG use the local projections method developed by Jordà (2005) to analyze the effects of fiscal shocks on different US variables. What distinguishes their approach from that used in previous papers is the use of data at a daily frequency. More commonly, data at lower, i.e. quarterly or yearly, frequency are used for fiscal policy analyses. Auerbach and Gorodnichenko (2016, p.177) argue that "[t]hese low frequencies can limit the plausibility of identifying assumptions (for example, due to delays between government spending announcements and their implementation) and reduce statistical power (for example, narrative shocks can account for only a few historical changes in fiscal variables)". However, daily data on government spending[3] is not easily obtainable. The authors create two variables on daily government spending, both referring to military

[3]Note that, government spending in this context means government consumption, not investment or transfers.

spending.[4] First, daily statements published by the U.S. Department of the Treasury are used to obtain information on daily payments to defense contractors, thus creating a variable on daily *actual spending*. Second, daily records on newly awarded (defense) contracts, which are published online by the U.S. Department of Defense (DoD) (now Department of War), are used to obtain a daily measure of government spending announcements, i.e. *future spending*. Both series include data from October, 6th 1994 to March, 28th 2014. As both spending series exhibit seasonal patterns, they are deseasonalized by a TBATS model[5] as presented by Livera, Hyndman, and Snyder (2011) before they are used for analysis. The deseasonalized series from the estimated TBATS model are then used in the following estimation equation:

$$X_{t+h} - X_{t-1} = \alpha_h G_t + \sum_{i=1}^{I} \beta_i \Delta G_{t-i} + \sum_{j=1}^{J} \gamma_j \Delta X_{t-j} + constant + error_{t+h}, \quad h = 0, ..., H \quad (1)$$

where X represents the outcome variable of interest (e.g. the exchange rate), G stands for deseasonalized government spending and Δ indicates the first difference operator. The left-hand side of the equation describes the change in X relative to previous periods. Lagged values of G and X control for potential serial correlation present in both values.[6] AG run regression (1) for each possible horizon h. Thereby they obtain a series of impulse responses α_h, which describe the effects of a change in G_t on X over a period of h business days. These responses are illustrated graphically as impulse response functions (IRFs). The direction of responses (positive or negative) and whether the results are statistically significant are most relevant for the interpretation of results.

Based on equation (1), AG examine the effect of a change in U.S. government spending on the exchange rate as well as other high frequency (financial) variables.[7] AG's main

[4]By using military spending, Auerbach and Gorodnichenko (2016, p. 178/9) want to "[..] minimize the possibility of reverse causality and other forms of endogeneity".

[5]The abbreviation stands for: **T**rigonometric seasonality, **B**ox-Cox transformation, **A**RMA errors, **T**rend and **S**easonal components.

[6]See Born, D'Ascanio, Müller, and Pfeifer (2021) for more details on the methodology and the underlying estimation procedure.

[7]The exchange rate is measured by the DTWEXM series from St. Louis Federal Reserve's (FED) FRED database, which displays the Nominal Major Currencies U.S. Dollar Index (Goods Only). Other variables of interest are: the TED Spread, CBOE DIJA Volatility Index, S&P 500 Stock Price Index, measures for Crude

finding is that the use of government spending data at a high frequency resolves the exchange rate puzzle previously found in the literature. When analyzing a shock in their variable on announced, i.e. future, U.S. military spending, the exchange rate reacts on impact and appreciates strongly and significantly. AG conclude that low-frequency spending data is not suitable for showing the response of high-frequency variables, such as the exchange rate and other financial variables, and thereby causes the commonly empirically observed depreciation of the exchange rate at odds with theoretical model predictions (also known as exchange rate puzzle). In line with theoretical arguments by Baxter and King (1993) and empirical findings by Ramey (2011), AG point out that it seems to be spending announcements rather than actual spending changes that determine macroeconomic responses to fiscal policy shocks. To explore the potential transmission channel through which the announcement spending series affects the economy further, AG look at the response of additional financial variables (see footnote 6). Although no statistical significance can be established for any of the financial variables considered, the directions of movement point towards a transmission through a demand channel, implying an increase in demand worldwide following the US spending shock. For instance, following the announcement shock, the gold price declines and there is an increase in interest rates and inflation expectations as well as foreign stock market indices. Liquidity and uncertainty measures on the other hand show no clear response in either direction, deeming other transmission channels (e.g. liquidity channel) less likely.

To evaluate the robustness of their spending measure and to assess whether their findings are driven by certain events in the sample, AG look at different subsamples. They expand their approach in equation (1) to allow for differences between recessions and expansions as well as periods when the zero lower bound on nominal interest rates was binding.

AG contribute to the fiscal policy literature in two ways. First they show that a fiscal shock leads to an appreciation of the exchange rate when using a high frequency series of military spending announcements, which is in line with theoretical models. Thereby they

Oil Prices, conventional gasoline prices, the gold fixing price, FTSE 100, Nikkei Stock Average, Nikkei 225 and S&P/TSX Composite Index.

offer a new explanation for the exchange rate puzzle found in the empirical fiscal policy literature. High frequency data on military spending announcements can provide a clear test of theory, which explains the deviation of theoretical and empirical predictions in the case of low frequency empirical studies. Second, they contribute to the discussion on fiscal shock identification by offering a new way of high-frequency identification and underlining the importance of announcement versus actual spending innovations.

The term exchange rate puzzle is used to describe a divergence between theoretical model predictions and empirical findings regarding the exchange rate's response to a fiscal policy shock. Most theoretical models predict that following an increase in government spending, the increase in domestic demand translates to a trade balance deficit through an increase in imports and a current account deficit, which in turn implies higher interest rates and therefore an increased demand for the domestic currency and hence an appreciation of the real exchange rate. Empirically, however, researchers have often found depreciations. For instance, Kim and Roubini (2008) and Monacelli and Perotti (2008) use vector autoregressive (VAR) models with Blanchard and Perotti (2002)-type shocks, respectively, and discover that in response to a fiscal policy shock in the US the real exchange rate depreciates. Ravn, Schmitt-Grohé, and Uribe (2012) analyse the effects of government spending on the real exchange rate (among other macroeconomic variables) for four countries and point out that a theoretical model with a deep habits mechanism[8] is able to explain the puzzling empirical finding. Other publications have pointed towards differences in country characteristics to explain the puzzle (e.g. Ilzetzki, Mendoza, and Végh (2013) or more recently, Miyamoto, Nguyen, and Sheremirov (2019)). Another explanation has been proposed by Corsetti, Meier, and Müller (2012), who coined the term "spending reversals". Empirically, positive government spending shocks are after some time followed by an undershooting due to self-correcting debt dynamics. Consumers anticipate these reversals and thus expect a fall in the long-term interest rate. These falling long-term rates are in turn associated with a depreciation of the real exchange rate. Born

[8]Deep habits are formed over individual goods rather than over aggregate consumption (Ravn, Schmitt-Grohé, and Uribe (2006)).

et al. (2021) point out that the direction of the fiscal shock matters as does the type of exchange rate regime. Based on a model with downward nominal wage rigidity, they show that a fiscal expansion in fixed exchange rate economies leads to an appreciation of the real exchange rate (which simultaneously mutes any output response), while a fiscal contraction lowers output but leaves the exchange rate unchanged.

Moreover, fiscal shock identification is an important topic in fiscal policy research generally as well as for the discussion on the exchange rate puzzle. Identification is important because government spending policy changes need to be exogenous, i.e orthogonal to other shocks (like states of the business cycle) in order to estimate effects empirically using time series methods. Common approaches to addressing this issue have been presented by Blanchard and Perotti (2002) or Ramey (2011), for example. In this context, a distinction between actual spending data and constructed "surprise" shocks is relevant. With regard to the exchange rate puzzle, Forni and Gambetti (2016) argue that it is important to distinguish between news, meaning anticipated, shocks, and surprise shocks. When accounting for fiscal foresight[9] through news shocks, Forni and Gambetti (2016) show that the exchange rate puzzle disappears and an appreciation is found empirically. They conclude that the puzzling depreciation applies only to surprise shocks. Ferrara, Metelli, Natoli, and Siena (2021) state that previous publications were unable to identify the fiscal shock correctly. The authors use narrative methods[10] to obtain a spending series, which they use in a proxy-SVAR model in their analysis and obtain empirical results in line with theory. They point out that the proxy-SVAR method should be preferred over other VAR methods, as the latter tend to react more strongly to structural changes in the way of how fiscal policy is conducted. This also leads to the puzzling depreciation of the exchange rate in other empirical papers.

AG add to this discussion, offering a new way to identify fiscal shocks based on daily spending data, which leads to an appreciation of the exchange rate in response to a spending

[9]A more detailed discussion on the implications of fiscal foresight for econometric analyses can be found in Leeper, Walker, and Yang (2013).

[10]Ferrara et al. (2021) build up on a narrative series constructed by Ramey (2011). The narrative series is obtained by reading news sources and from that constructing a measure for (military) spending.

shocks. AG argue that the main reason for the depreciation puzzle is an artifact of inadequate identification methods and insufficient data used in previous empirical exercises, especially when considering the impact on fast moving variables. Auerbach and Gorodnichenko (2016, p.179) conclude that it is the "mistiming of shocks in previous papers and their use of actual spending rather than news about spending" that explains the different results.

The code and data used to obtain their results are made publicly available by AG. I base my reproduction fully on their supplementary material. The results are presented in the next section.

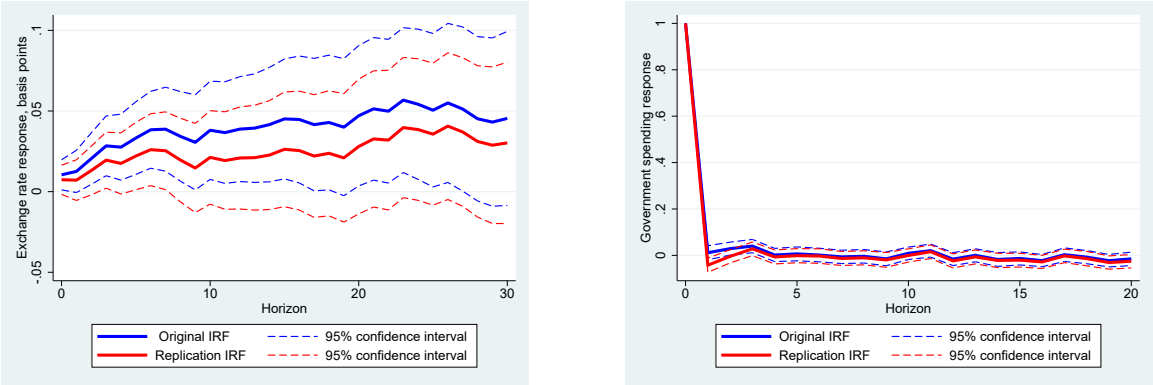
3 Replication

The focus of my replication is on recreating AG's main findings as presented in the paper. The authors provide a full set of replication materials[11]online, which are used without modification, and no additional code is introduced. The accompanying documentation is highly detailed. AG include the code and data files necessary to reproduce the figures of the publication as well as the raw data files and show the steps that are necessary to clean the data. I do not have access to the webscraping algorithm used by AG, but the raw data from webscraping is included in the material. This is therefore the starting point for my replication. The authors also add their cleaned data file, called `step006`. A full overview of all replication materials provided can be found in Table A1 in the Appendix. One piece of relevant information that is missing from those files is the software versions used, making it impossible to check whether the unknown software versions contributed or caused the mismatches described below.

A main finding by AG is that changes in the exchange rate and other high frequency variables can be captured appropriately through the use of high frequency data on future government spending, meaning announced spending in comparison to actual spending. Consequently, the response of the exchange rate following a fiscal shock is no longer puzzling but

[11]The full supplementary material can be found at: `https://link.springer.com/article/10.1057/imfer.2015.15`.

in line with what is expected by theory. AG illustrate this finding in Panel A of Figure 4 (Auerbach and Gorodnichenko (2016, p.192)), which illustrates the exchange rate's response following a shock to the spending announcement series [12]. Figure 2 below illustrates the differences between my replication of Figure 4, Panel A and the original results. Based on the cleaned data file provided by AG, I am able to generate the IRF presented in the original publication. This IRF is displayed in blue in Figure 2, while the IRF from my replicated file is displayed in red. I find - similar to AG - an appreciation of the exchange rate measure, but the replication IRF is flatter than in the original publication. While AG report a peak response of 0.0568 (at around 25 working days), my peak is at 0.0406. Additionally, my results are no longer statistically significant for the longest part of the series.



(a) Real Exchange Rate **(b)** Government spending

Figure 2: Impulse response functions (IRFs) to a unit shock in the DoD announcement government spending series for 1) the U.S. real exchange rate on the left and 2) government spending on the right. The x-axis (horizon) displays days. Red lines show the results from my replication, blue lines display the original results by AG.

The response of the DoD announcement variable to a unit shock (Panel B) in my case is slightly negative following the impact response.

The replication of Figure 4 from the original publication serves as an example here. After

[12]The exchange rate in this case is in log volumes, the DoD announcement series measures the daily log volume of deseasonlized and detrended contracts awarded by the DoD, and deterministic trends are included in the estimation equation.

attempting to replicate the study, I find that I am unable to exactly replicate most of the figures AG provided. More specifically, the only figure I successfully reproduce is Figure 1 of the original publication, which depicts daily data on actual government spending (based on daily statements by the U.S. Treasury; Auerbach and Gorodnichenko (2016, p.182)). For all other figures, similar trends are identified but no exact replication is possible. Importantly, this does not mean that I contradict AG's findings. Instead, my findings support the general results presented in the original publication, they are merely less pronounced and do not show statistical significance in my replication. The variables of interest still move in the same direction throughout. All other replicated figures can be found in subsection A.4 in the Appendix.

Given the novelty of AG's approach, it is reasonable to examine the replicability of their results. In this context, it is important to identify the reasons for the deviation between the original and replicated results. The source of divergences most likely lies within the data cleaning process. In this process, raw data files are combined and then deseasonalized.[13] AG provide both the "cleaned" data file `step006` and the raw data files necessary to generate `step006` using their code. After running the code to generate the file `step006` myself, the generated file differs from the provided data file by the authors. As a result, I am unable to generate the same figures presented in the paper given the newly generated `step006`. Interestingly, when using the cleaned `step006` file provided by AG, the figures of the paper can be exactly replicated with the code provided.[14] I therefore conclude that relevant causes for deviations between my and AG's results are to be found within the files that clean the data. Figure 3 illustrates this observation.

[13]The data cleaning process can roughly be separated into three stages. First, Stata files `step000` to `step004` are used to combine data from different sources and align dates, second, the military spending data gets deseasonalized. This process is done in R. Third, Stata files `step005` to `step007` then combine the deseasionalized data with the prior files and bring them into the right format for the regression analysis.

[14]The only exception here is Figure 2 (showing new contracts awarded by the Department of Defense in millions of dollars, see A.4 in the Appendix), which differs from the published version, though showing only minor deviations.



Figure 3: Illustration of problems within replication. Step000 - step007 refer to different Stata do-files provided by AG, the R-File corresponds to the R Script in the replication material which deseasonalizes the spending series.

On a closer look, I find that the discrepancy between my `step006` file and that by AG can be attributed mostly to differences in the two spending series. The correlation coefficient between the original and the replicated spending announcement series is 0.957 and for the actual spending series it is 0.937.[15] As the spending series are crucial for all following analyses, this small difference can be responsible for the slight deviation of my figures from the original. Besides, autocorrelations for the spending announcement series in the original data set are only positive while my data displays negative and positive autocorrelations – however never bigger than 0.04 (the lag length is chosen at 5). Based on this finding, I identify three main problems with the data cleaning process as potential reasons for deviation.

First, one potential source can be found within the R file provided by the authors. This part of the code describes how to deseasonalize the daily spending data series. A comparison of the replicated `step006` file with the file provided by AG shows some differences within the variables generated through this step. A main challenge when using high frequency spending data is to fully separate anticipated from unanticipated changes to spending as their respective effects can differ. Auerbach and Gorodnichenko (2016, p. 184) point out that "[...] the daily data exhibit predictable movements on certain days of the week, days/weeks of the month, and months of the year, so one may observe systematic cycles at a variety of

[15] All other variables, which are unrelated to the spending series and are included in the cleaned data file are identical in both `step006` files.

frequencies." Consequently, to account for these seasonal components the authors apply a TBATS model to their spending data by using the corresponding function in R.[16]

Before running the R script, AG drop all other variables besides the DoD announcement series, the Treasury data series on actual spending, and their real exchange rate measure as well as a date variable. Thus, the file that is loaded into R only contains four variables.[17] After importing the data into R and adjusting directory names to match mine, R shows an error message when trying to run the code. The reason for the error message is easily identified: in the R script TBATS is applied to six different variables. Given that only three variables are accessible through the dataset that has been loaded into R, I simply do not have enough data to match the functions in the script. In order to run the R script, I therefore comment out the lines of code that refer to non-existing variables.[18] I can do so because each variable is included in a different function. After doing so, I am able to apply TBATS to both spending variables (OLITFWDVP and OLIDC2) and deseasonalize the two relevant series. The other variables that were commented out do not appear in other and later steps of the replication material. Hence, while this shows problems with the provided R file, commenting out parts of the code should not lead to issues in the downstream replication effort.

When comparing my deseasonalized spending series with those used by AG[19] I do not obtain the same values. Table A2 in the Appendix illustrates these differences. I find that the seasonal periods accounted for by TBATS in my R script do match the description in the original paper. Auerbach and Gorodnichenko (2016, p.185) state that "[a]fter extensive specification searches, we include four cycles with periods that correspond to weekly, fortnightly, monthly, and yearly durations". However, the file available to me contains only 3 seasonal components.[20] I conclude that the provided R script is not the one

[16] `https://www.rdocumentation.org/packages/forecast/versions/8.4/topics/tbats`.

[17] These are named DATE, OLITFWDVP, OLIDC2, LDTWEXM.

[18] A detailed description of these steps can be found in the replication material provided with this paper.

[19] Table A1 in the original paper displays descriptive statistics of the relevant series.

[20] In the TBATS function in R those are specified as `seasonal.periods=c(5,21.7,260.8)`.. The authors consider only working days, so that `5` probably corresponds to the weekly duration, while `21.7` and `260.8` could reasonably correspond to monthly and yearly durations, respectively. In this case, a seasonal component included accounting for a fortnightly duration is missing.

that has been used by the authors to obtain the results in the paper. Nonetheless, other explanations remain possible. Unfortunately, an email inquiry to the original authors has remained unanswered.

Second, there are also deviations in the non-deseasonalized data. I find that the replicated spending announcement series (DC2) already differs slightly from the one by AG before being loaded into R, thus before deseasonalization. More specifically, I find 63 mismatches between my spending announcements (DC2) variable and the original. At the same time, my actual spending series (OLITFWDVP) is identical to AG's - which is illustrated in Figure 1 in the original publication and which I can replicate exactly. I cannot find an explanation for the small deviation in the DC2 variable.

Third, there is a file missing from the replication material which is used in the do-file `step005`. AG use a file called KalmanDoD within a joinby command (thus, a horizontal pairwise merge). This file is not included in the material provided. However, the variables that are added at this point are not used again at any other point of the analysis. Therefore, the missing file should not have any effects for replicability.[21]

To conclude, I am able to replicate the general trends identified by AG but am not able to establish a similar statistical significance and exactly reproduce the original results. I attribute the discrepancies mainly to problems within the R file, which is used to deseasonalize the spending serie as the seasonal adjustment periods described in the paper do not match the seasonal periods in the R file. Additionally, there are small differences with respect to the raw data files.

[21]Additionally, a few more changes were required in order to run all parts of the replication material smoothly. These changes should not be responsible for any deviations and I only list them for completeness here. I have to change the Stata matrix size limit from 8000 to 800 as my Stata version does not support a bigger size. However, no error messages result from this change. Moreover, the file loaded into R needs to be saved in Stata12 or lower (in step004). I assume the original authors used Stata version 12 file and specify `version(12)` with the saveold command. Additionally, when loading the csv file saved from R containing the deseasionalized series back into Stata, I have to add `varnames(1)` for Stata to identify the variables and proceed correctly. Additionally, in step006 logged first difference variables are created for a set of variables. There is one variable (ADS) which does not show up in the data file and therefore needs to be commented out.

4 Extension

To further evaluate the results by AG, I extend the spending series with additional data from July 2014 to December 2019. My replicated findings above support the results by AG but are less pronounced and not statistically significant. I hope to improve on the latter by using a longer time series and thereby evaluating whether the significance can be improved by increased power and more data.

I am most interested in the robustness of AGs main finding regarding the exchange rate puzzle. As mentioned earlier, AG's main finding is that using high frequency data on spending announcements fiscal shocks leads to an empirical appreciation of the exchange rate in line with theoretical predictions. I therefore focus on thisö finding (thus, Figure 4 of the original publication) and extend the respective data series on DoD spending announcements and the exchange rate.

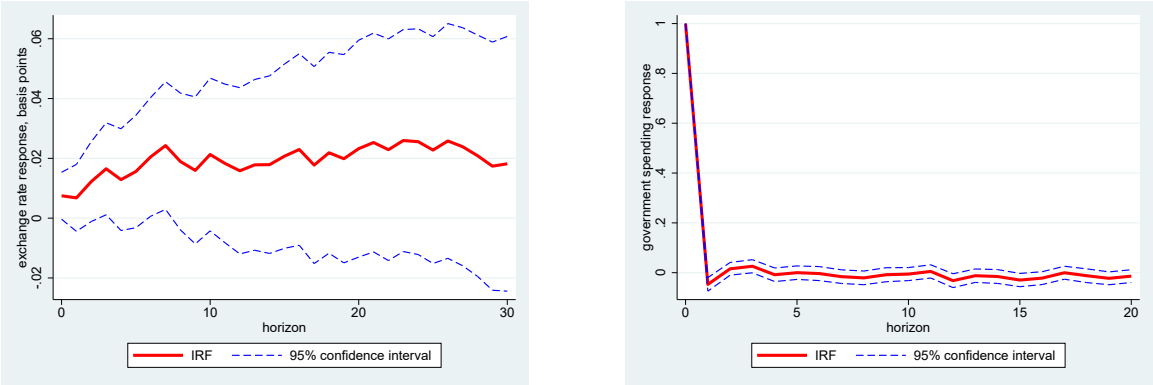
I take AG's webscraped data as given, and follow the description by AG in their paper to retrieve data on newly announced contracts by the DoD from their website (see `https://www.defense.gov/News/Contracts/`.) Unfortunately, the announcement data is not available in a format that is easily usable for statistical purposes.[22] Like AG, I therefore use a web-scraping algorithm to extract the respective data[23] I obtain data from July, 1st 2014 to December, 20th 2022. This is the earliest start date possible as the DoD data archive does not include earlier entries. AG use data from 1994, October 6th to 2014, July 14th, implying that there must have been an adjustment in the publication dates by the DoD. Luckily, the available data is sufficient to extend the spending announcement series. Unfortunately, this limits the opportunities to compare my web-scraped series with that by AG to confirm consistency of my web-scraped data. I find that for the overlapping dates, my and AG's data

[22]More specifically, the data is provided as text within written paragraphs. For more information see the description on pages 183/4 in AG.

[23]The web-scraper is included in my replication material. I do not have access to the algorithm used by AG and therefore fully rely on the description provided in their publication and within the code files. I define the necessary criteria for the algorithm myself and then generate the web-scraping algorithm with help from a computer scientist.

points are sufficiently close.[24] From a comparison of my web-scraped information for dates after 2014, July 14th (on numerous days chosen at random) and the DoD announcements on the website I conclude that my extracted information is reasonably correct. I am therefore able to combine my data with AG's series and construct an extended data series on daily spending announcements by the DoD ranging from 1994 to 2022. Figure A5 in the Appendix displays the extended data series.

Regarding the exchange rate series, I find that it has been discontinued at the end of 2019. I therefore decide to only extend the sample period up until the end of 2019. With this information and the replication material, I am then able to analyze the effects of a fiscal shock to the (extended) spending announcement series regarding the response of the exchange rate. Figure 4 illustrates the results obtained with the extended data series.



(a) Real Exchange Rate **(b)** Government spending

Figure 4: IRFs to a unit shock in the DoD announcement government spending series for an extended sample dating from 1994 to 2019.

The response of the exchange rate looks similar to the one I obtained on the shorter sample. I still find a positive response, indicating an appreciation following a government spending announcement shock. This is in line with theoretical predictions. In comparison to Figure 2, the results reported here are flatter than before, but generally comparable to those from my replication. Hence, more data does not lead to a more significant result. However, it is not

[24]There are slight deviations in individual contracts, but on a daily level the amounts obtained through my webscraping algorithm are comparable to that by AG.

clear whether this effect arises because of errors in the code provided or because the effect is not as strong as hoped.

5 Conclusion

This paper reports the results from a replication of AG. While I am cannot fully replicate the original results, the direction of the results is identical, albeit less pronounced. Additionally, an extension of the sample period does not does not increase the statistical significance of the estimated effects. Again, the direction of the effects remains unchanged but remain statistically insignificant. I conclude that my replication and extension exercises fail to provide further proof for the explanation offered by AG while simultaneously not contradicting AG's findings with respect to their methodology and results. Even though the focus of this paper is on a single publication, it should not be misread as a criticism of the publication by AG. Instead my findings should be understood as part of a bigger discussion on reproducibility and replicability in economics.

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Appendix

A.1 Replication material - Overview

Name	Description	File type
`step000` - process DoD data - 001	Imports DoD data on announced contracts and cleans them	Stata do-file
`step001` - process Haver data	Imports and merges data series on cash deposits to and cash withdrawals from the Federal Reserve Account as well as cash withdrawals for defense vendor payments	Stata do-file
`step002` – process FRED data	Imports a wide variety of macroeconomic data series downloaded from the FRED database (mostly financial variables and components of GDP) at a daily frequency and aggregates them to a monthly frequency	Stata do-file
`step003` – combine DoD, Haver and FRED	Combines all dataset from the previous steps and generates dummies for holidays	Stata do-file
`step004` – prepare file for R with date alignment	controls for holidays and adjusts both spending series so they can be get deseasonalized in R	Stata do-file
Script R	file which applies the TBATS function to the spending series	R Script
`step005` – read file from R and merge with aligned data	merge file from step003 with the deseasonalized data (from R); add additional data (KalmanDoD)	Stata do-file
`step006` – prepare data for regressions	generates logged first difference versions and first differenced variables; saves resulting file as step006	Stata do-file
`step007` – additional stock market data	imports additional stock market data and adjusts date format	Stata do-file
`step3xx`	create figures in the main text; one do-file for each file	Stata do-files
`step4xx`	create figures in the Appendix; one do-file for each file	Stata do-files
`step500`	creates Table A1 in the Appendix	Stata do-files
runme	lists all do-files needed to create figures in the main text and appendix; uses the provided cleaned dta file step006	Stata do-file

Table A1: This table displays the different files included in the folder "code" provided by AG in their supplementary material. The code is structured in different steps. `Step000 - step007` prepare and deseasonalize the data; all other files are needed to recreate the figures in the paper. The replication material includes three more folders. The "raw data" folder consists of the raw data files, which are mostly used in `step000 - step002` above. Additional data files are saved in the folder "workfiles", which also includes AG's cleaned data file `step006`, that allows to exactly replicate the figures in the original paper.

A.2 Descriptive Statistics

	DoD announcements		Treasury statements	
	AG	Replication	AG	Replication
Observations	4895	4895	4587	4587
Mean	−0.001	−0.020	−0.004	−0.006
Standard Deviation	1.355	1.391	0.315	0.308
Interquartile range	1.695	1.767	0.327	0.318
1st Percentile	−3.028	−3.173	−1.073	−0.966
5th Percentile	−2.242	−2.303	−0.572	−0.512
10th Percentile	−1.673	−1.758	−0.331	−0.338
25th Percentile	−0.863	−0.919	−0.143	−0.150
50th Percentile	−0.071	−0.061	0.024	0.010
75th Percentile	0.832	0.848	0.184	0.168
90th Precentile	1.731	1.732	0.336	0.328
99th Percentile	3.525	3.622	0.619	0.711

Table A2: This table corresponds to Table A1 in Auerbach and Gorodnichenko (2016, p.214). It presents descriptive statistics for the two *deseasonalized* government spending series. TDC2 corresponds to the series of DoD announcements and DVP denotes daily statements by the U.S. Treasury.

A.3 Extended Data Series on DoD Spending Announcements

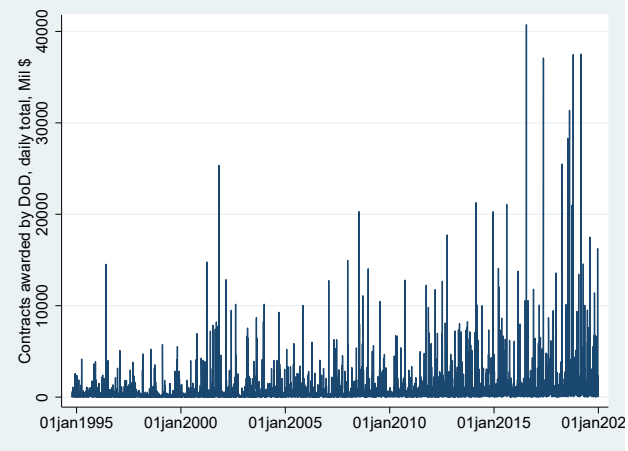


Figure A5: This figure corresponds to Figure 2 in Auerbach and Gorodnichenko (2016, p.185). It shows the newly announced contracts by the Department of Defense at a daily frequency in millions of dollars from January 1995 to May 2022.

A.4 Comparison of original and replicated figures

This section reports the figures constructed from the original data file `step006` provided by AG (on the left-hand side, title: Original) and compares them to the ones obtained when using the replicated `step006` file (on the right-hand side, title: Replication). The original data file constructs the Figures that are shown in the paper with one exception. Figure 2 does not seem to be the same as reported in Auerbach and Gorodnichenko (2016, p.185). I only report a comparison of the figures in the main text. A comparison of figures from the Appendix of AG is available upon request.

Figure 1. Daily Measures of Government Spending. Panel A: Daily Cash Withdrawals from Fed. Res. Acct. for Defense Vendor Payments; Panel B: Quarterly totals

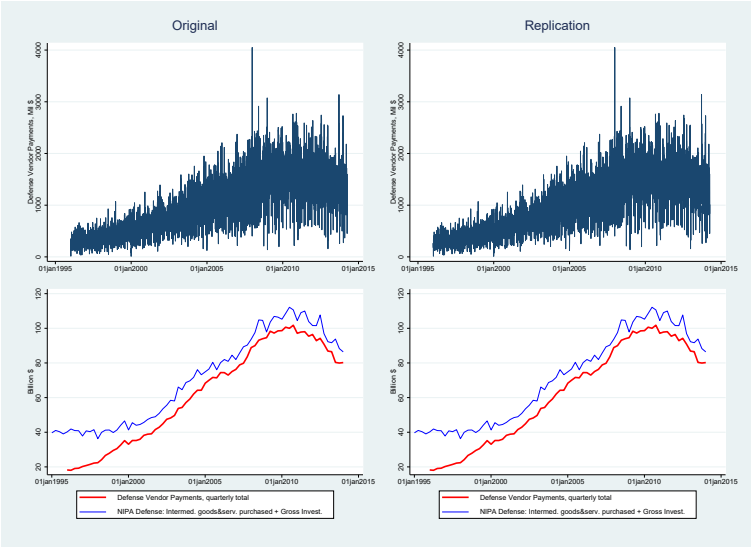


Figure A6: This figure shows the differences between the original (left) and replication (right) results regarding Figure 1 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 2. New Contracts Awarded by the Department of Defense in Millions of Dollars. Panel A: Daily totals; Panel B: Monthly totals

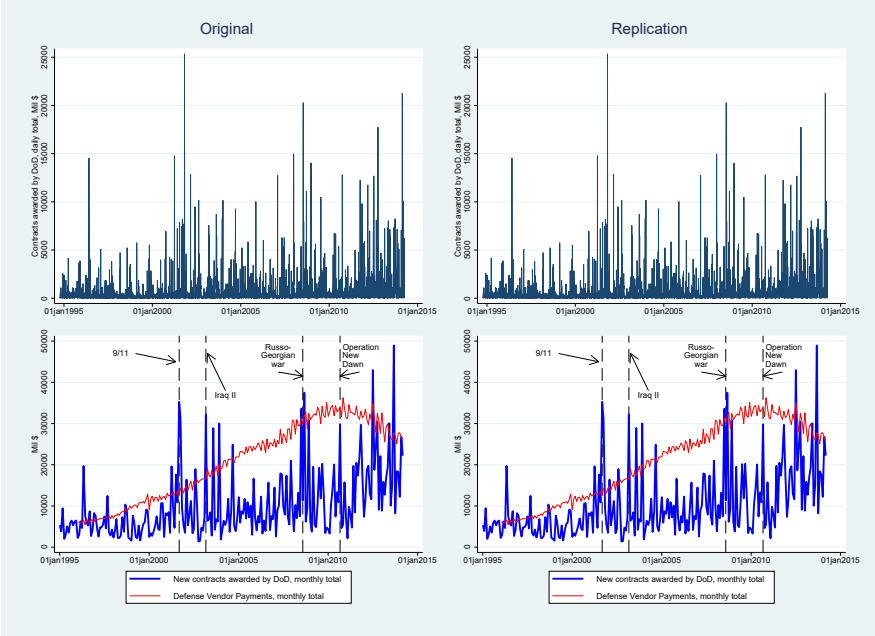


Figure A7: This figure shows the differences between the original and replication results regarding Figure 2 in AG. Interestingly, even the original file `step006` does not exactly replicate the figure shown on p.185 in the paper. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 3. Response of Index of Stock Prices of Defense Corporations to a Government Spending Shock. Panel A: Response to DoD announcements; Panel B: Responses to a shock in daily payments to defense contractors (U.S. Treasury daily statements)

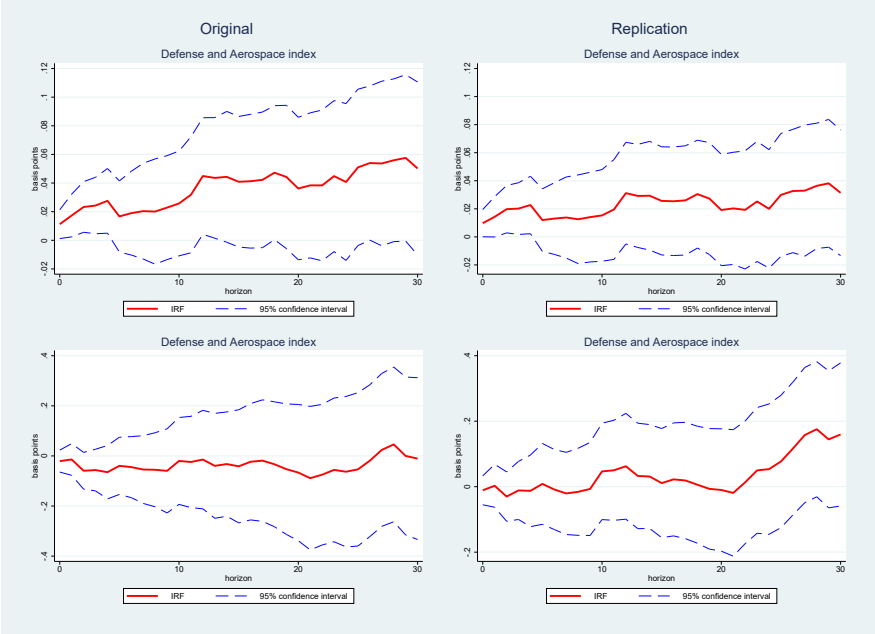


Figure A8: This figure shows the differences between the original (left) and replication (right) results regarding Figure 3 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 4. Impulse Responses to a Government Spending Shock, Daily Data. Panel A: Department of Defense announcements



Figure A9: This figure shows the differences between the original (left) and replication (right) results regarding Figure 4 Panel A in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 4. Impulse Responses to a Government Spending Shock, Daily Data. Panel B: Daily payments to defense contractors

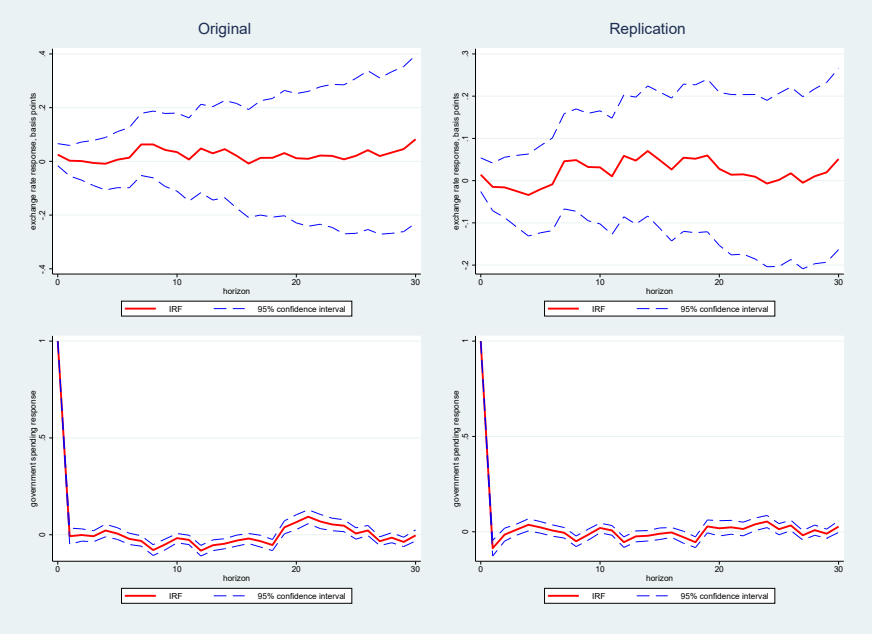


Figure A10: This figure shows the differences between the original (left) and replication (right) results regarding Figure 4 Panel B in AG. The figure title is copied from the original figure to give some context. For further explanations see AG.

Figure 5. Impulse Response to a Government Spending Shock, Monthly Data. Panel A: Department of Defense announcements

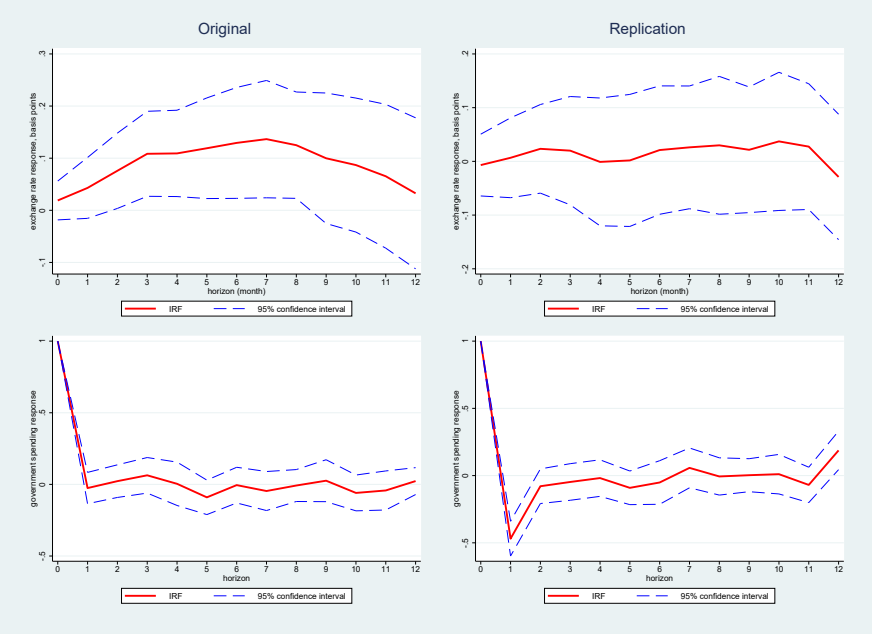


Figure A11: This figure shows the differences between the original (left) and replication (right) results regarding Figure 5 Panel A in AG. The figure title is copied from the original figure to give some context. For further explanations see AG.

Figure 5. Impulse Response to a Government Spending Shock, Monthly Data. Panel B: Daily payments to defense contractors

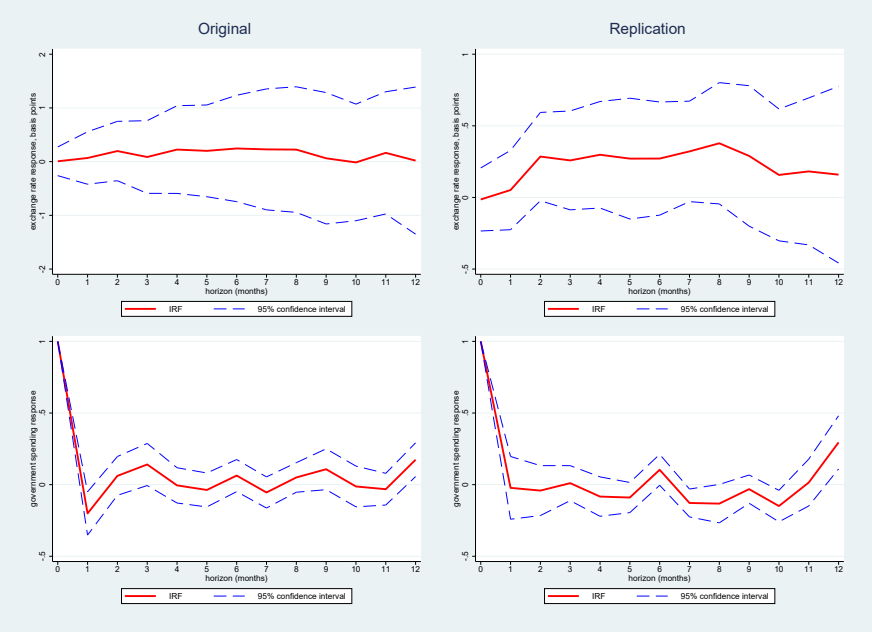


Figure A12: This figure shows the differences between the original (left) and replication (right) results regarding Figure 5 Panel B in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 6. Responses of Additional Macroeconomic Variables to a Government Spending Shock



Figure A13: This figure shows the differences between the original (left) and replication (right) results regarding the first row of Figure 6 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 6. Responses of Additional Macroeconomic Variables to a Government Spending Shock



Figure A14: This figure shows the differences between the original (left) and replication (right) results regarding the second row of Figure 6 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 6. Responses of Additional Macroeconomic Variables to a Government Spending Shock

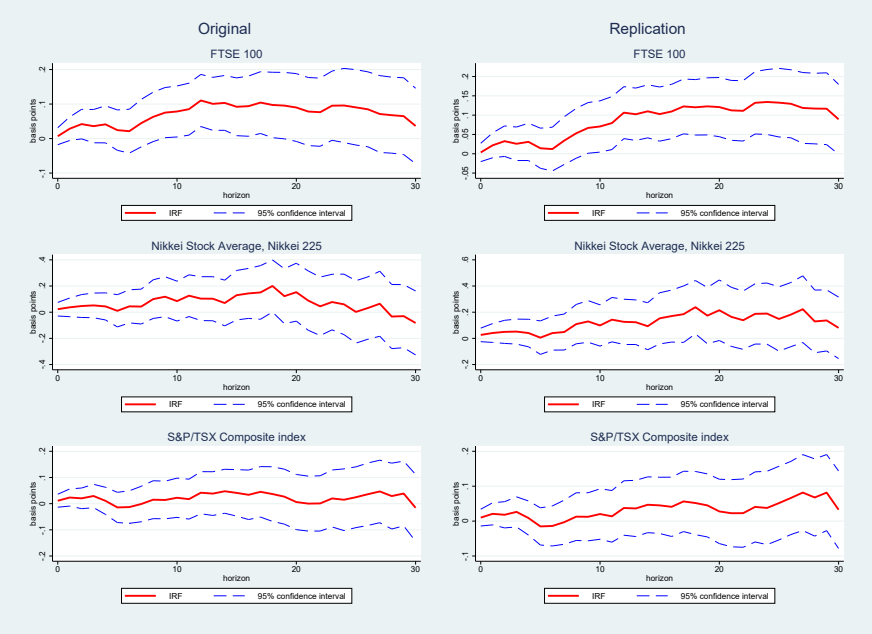


Figure A15: This figure shows the differences between the original (left) and replication (right) results regarding the third row of Figure 6 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 7. Responses of Interest Rates and Inflation Expectations to a Government Spending Shock

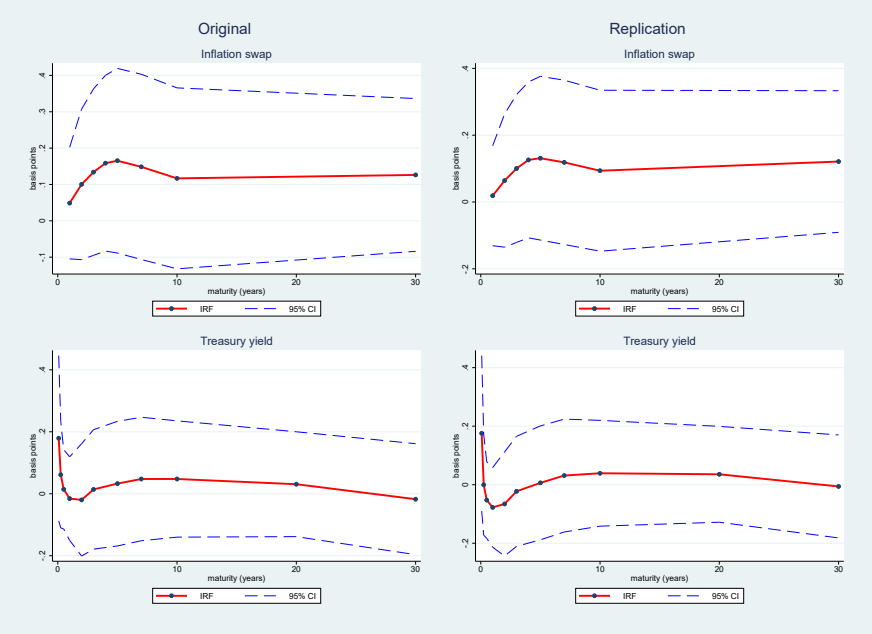


Figure A16: This figure shows the differences between the original (left) and replication (right) results regarding the first row of Figure 7 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 7. Responses of Interest Rates and Inflation Expectations to a Government Spending Shock

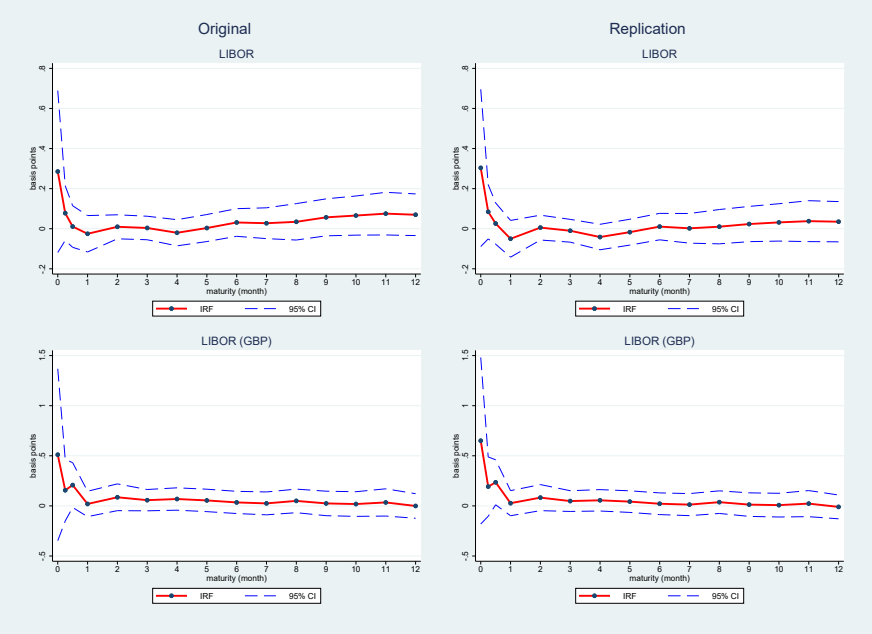


Figure A17: This figure shows the differences between the original (left) and replication (right) results regarding the second row of Figure 7 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 8. Stability of the Responses to a Government Spending Shock. Panel A: Pre-Great Recession period; Panel B. Period with the binding zero lower bound on the nominal interest rates; Panel C. Period with the binding zero lower bound on the nominal interest rates, control for TED spread.



Figure A18: This figure shows the differences between the original (left) and replication (right) results regarding Figure 8 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 9. Rolling Regressions, 24-Month Window

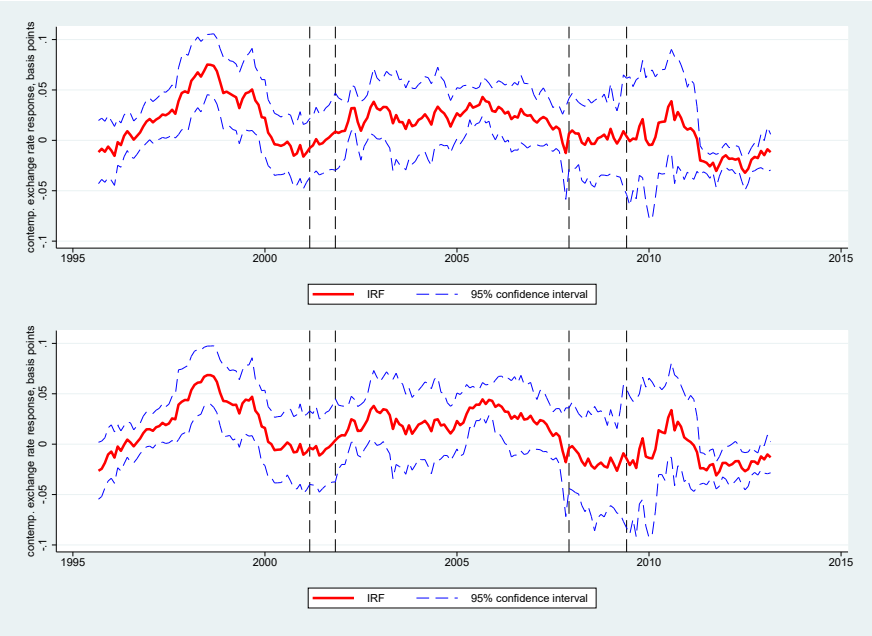


Figure A19: This figure shows the differences between the original (left) and replication (right) results regarding Figure 9 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 10. State-Dependent Responses to a Government Spending Shock, Business Cycle. Panel A: Expansion vs. Recession

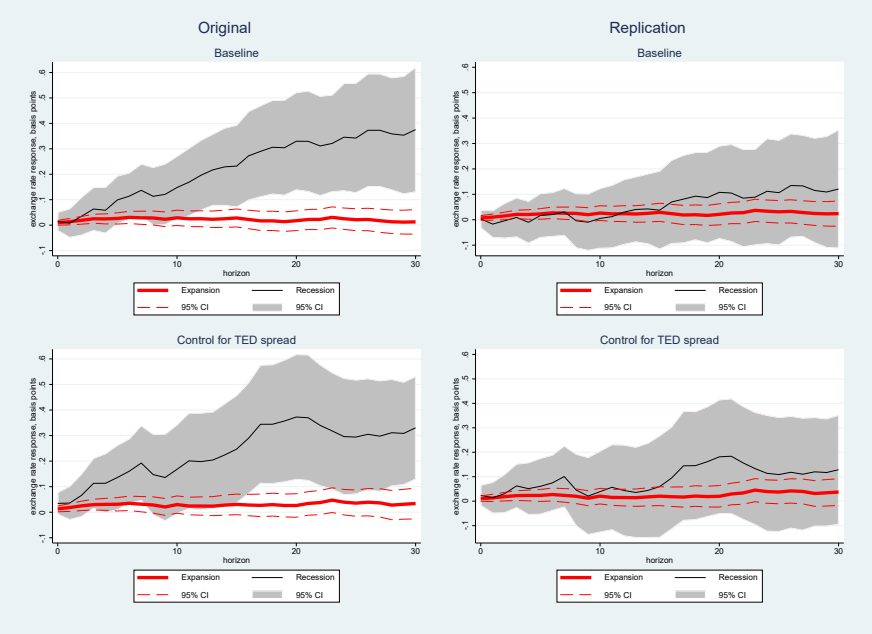


Figure A20: This figure shows the differences between the original (left) and replication (right) results regarding Panel A of Figure 10 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 10. State-Dependent Responses to a Government Spending Shock, Business Cycle. Panel B: Expansion vs. Recession by ZLB regime

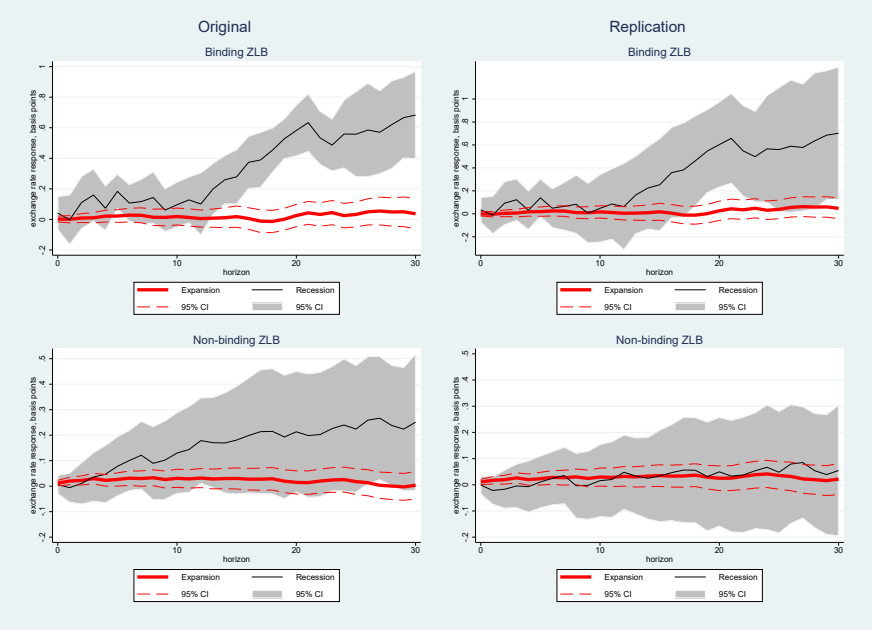


Figure A21: This figure shows the differences between the original (left) and replication (right) results regarding Panel B of Figure 10 in AG. The title is copied from the original figure to give some context. For further explanations see AG.

Figure 10. State-Dependent Responses to a Government Spending Shock, Business Cycle. Panel C: Expansion vs. Recession by ZLB regime, control for the TED rate

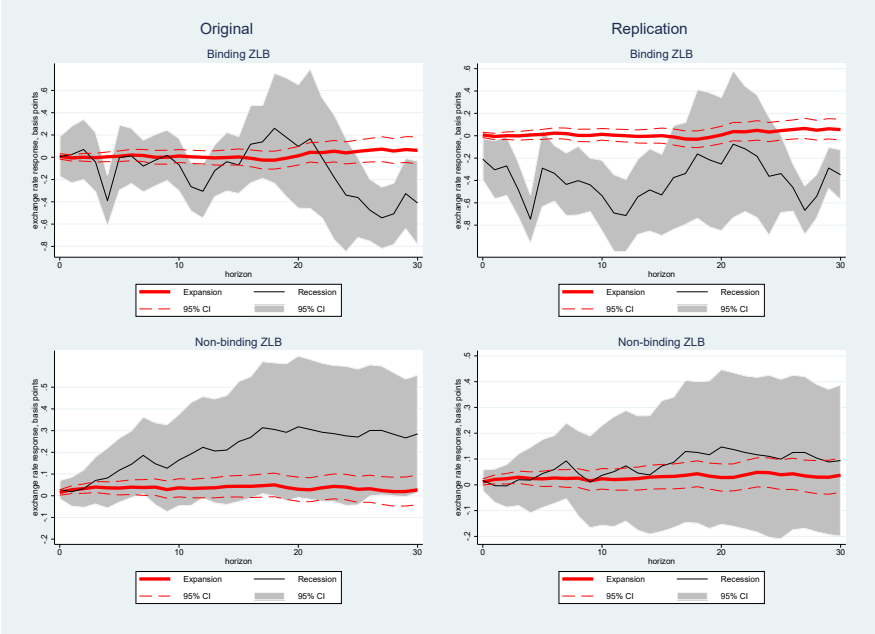


Figure A22: This figure shows the differences between the original (left) and replication (right) results regarding Panel C of Figure 10 in AG. The title is copied from the original figure to give some context. For further explanations see AG.